

✓ Scenario 1: Selecting Key Features for Diabetes Prediction

□ **Machine Learning Concept:** Feature Selection

□ **Question:**

A healthcare startup wants to build a model to predict the onset of diabetes. The dataset has 50 features, but the team wants to use only the most relevant ones. How can they apply feature selection effectively?

□ **Steps to Achieve:**

- Collect data from patient health records (e.g., BMI, glucose, age, family history, etc.).
 - Preprocess the data: handle missing values and scale features.
 - Use correlation heatmaps to eliminate highly correlated features.
 - Apply *SelectKBest* with mutual information to identify the top features.
 - Optionally, apply Recursive Feature Elimination (RFE) with Logistic Regression.
 - Train the model using selected features.
 - Evaluate using Accuracy, ROC-AUC, and F1-Score.
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✓ Scenario 2: Start a New Django Project

□ **Question:**

Start a New Django Project: You're creating a new project for managing student data. The project name is student_portal. What is the step-by-step command to start it? write logic for this django project.

Answer:

```
django-admin startproject student_portal
```

□ **Step-by-Step Explanation:**

- Open your terminal or command prompt.
- Navigate to the directory where you want your project.

- Run the command:
→ `django-admin startproject student_portal`
 - This creates a folder named `student_portal` with default Django files.
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✓✓Scenario 3: Feature Reduction in a High-Dimensional Marketing Dataset

☐ **Machine Learning Concept:** Feature Selection

☐ **Question:**

A company is analyzing customer purchase patterns with 200+ behavioral features. How can they reduce dimensionality without losing predictive power?

☐ **Steps to Achieve:**

- Collect and preprocess data (normalize values, handle missing data).
 - Use PCA (Principal Component Analysis) to reduce dimensionality.
 - Alternatively, apply embedded methods like Lasso Regression for feature elimination.
 - Retain top components/features that explain most variance.
 - Train the final model using these components.
 - Evaluate using RMSE (if regression) or Accuracy (if classification).
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✓✓Scenario 4: Personalized Book Recommendation System

☐ **Machine Learning Concept:** Collaborative Filtering

☐ **Question:**

A digital library wants to recommend books to readers based on what similar readers liked. How should they design this system?

☐ **Steps to Achieve:**

- Build a user-item interaction matrix from user ratings.

- Calculate similarity between users (user-based CF) or books (item-based CF) using cosine similarity or Pearson correlation.
 - Recommend books that similar users liked but the target user hasn't read yet.
 - Handle sparsity using matrix factorization or KNN.
 - Evaluate using RMSE and Recall@K.
 - Update recommendations as new ratings come in.
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✓ Scenario 5: Feature Selection for Credit Risk Assessment

☐ **Machine Learning Concept:** Feature Selection

☐ **Question:**

A bank wants to assess the risk level of credit applicants using only the most important financial indicators. How can they reduce the number of features?

☐ **Steps to Achieve:**

- Collect applicant data: income, credit score, employment length, etc.
 - Preprocess: handle missing values, encode categoricals.
 - Use tree-based models like Random Forest to compute feature importance.
 - Drop low-importance features.
 - Validate the reduced feature set with a classifier (e.g., Decision Tree).
 - Evaluate using ROC-AUC, Precision, and Recall.
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✓ Scenario 6: News Article Recommendation Based on Reading Behavior

☐ **Machine Learning Concept:** Hybrid Recommendation System

☐ **Question:**

A news app wants to recommend articles based on both article similarity and user reading history. How can they implement a hybrid system?

❑ **Steps to Achieve:**

- Collect user reading behavior and article metadata (topics, source, etc.).
 - Build a content-based recommender using TF-IDF on article content.
 - Build collaborative filtering using user-article interaction matrix.
 - Blend both recommenders using a weighted average of similarity scores.
 - Evaluate using CTR (Click-through rate) and Time Spent.
 - Continuously retrain as new users and articles arrive.
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✔ Scenario 7: Feature Selection in Spam Email Detection

❑ **Machine Learning Concept:** Feature Selection

❑ **Question:**

You're building a spam detection model and have thousands of text features from emails. How do you identify the most useful ones?

❑ **Steps to Achieve:**

- Convert emails into TF-IDF vectors or word counts.
 - Apply feature selection using chi-square test to select words most correlated with the label (spam or not spam).
 - Optionally use embedded models like Lasso to reduce features.
 - Train a classification model on selected features.
 - Evaluate with Precision, Recall, and F1-Score.
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✔ Scenario 8: Recommendation System for Online Learning Platform

❑ **Machine Learning Concept:** Collaborative Filtering

❑ **Question:**

An ed-tech platform wants to recommend courses based on what similar learners have enrolled in. What steps would you take?

☐ **Steps to Achieve:**

- Build a user-course interaction matrix.
 - Use collaborative filtering (matrix factorization like SVD or KNN) to find similar learners or courses.
 - Recommend courses highly rated by similar users.
 - Handle cold-start using popularity or content-based filtering.
 - Evaluate with RMSE, Recall@K, and engagement rate.
 - Update recommendations periodically as user behavior changes.
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✓ **Scenario 9: Identifying Key Features for Predicting Car Prices**

☐ **Machine Learning Concept:** Feature Selection

☐ **Question:**

You're developing a car price prediction tool. With 100+ features (e.g., brand, mileage, engine type), how do you reduce complexity?

☐ **Steps to Achieve:**

- Collect car listings and prices.
 - Encode categorical features and normalize numerical ones.
 - Apply correlation analysis to drop redundant features.
 - Use feature importance from Gradient Boosting to rank features.
 - Retain top 15–20 features and retrain the model.
 - Evaluate using RMSE and R^2 score.
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✓Scenario 10: Cold-Start Recommendation for New Users in a Shopping App

□ **Machine Learning Concept:** Recommendation System

□ **Question:**

How do you recommend products to new users who haven't interacted with anything yet?

□ **Steps to Achieve:**

- Use popularity-based recommendations (e.g., top trending or top-rated items).
- If user demographic info is available, recommend based on age/gender/location using filtered subsets.
- Use content-based filtering to match products to user's registration interests (if provided).
- Gradually switch to collaborative filtering once the user starts interacting.
- Evaluate based on user engagement and product click rate.